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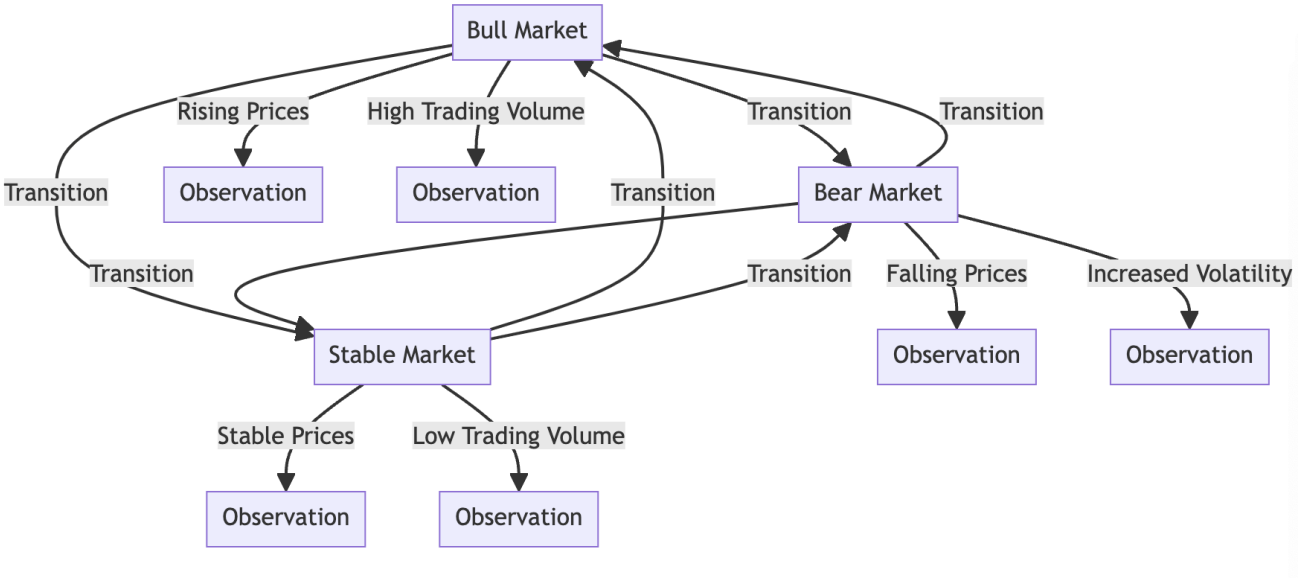
DynamicBayesianTrading

Intuition

- The behavior of the stock market is changing overtime, it's not a stationary process. Approach: Using a temporal model (a model that evolves over time)
- The suitable model(hierarchical hidden markov model)



The relationship between a HMM and DBN

A Hidden Markov Model (HMM) is a special type of Bayesian Network (BN) called a Dynamic Bayesian Network (DNB). A HMM may be represented in either matrix form for computation for as a graph for understanding the states and transitions. A DBN is a BN used to model time series data and can be used to model a HMM.

We can use the same parameter estimation as DBN such as EM algorithm.

The diagram of the model

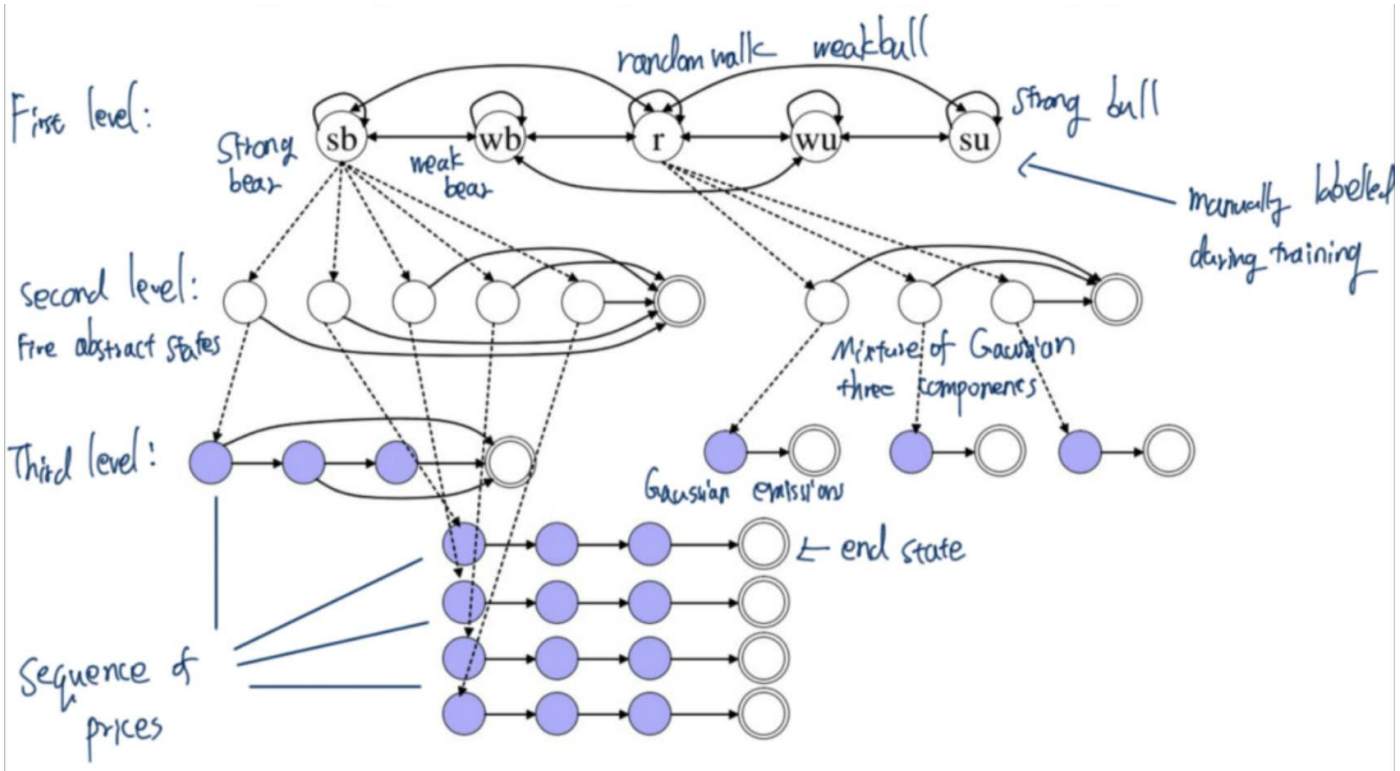


Fig. 1. State transition diagram for HHMM modelling the trend patterns of stock prices. Gray nodes are observation nodes, double circles are end states, and white circles are hidden states.

Stock Trading by Modelling Price Trend with Dynamic Bayesian Networks

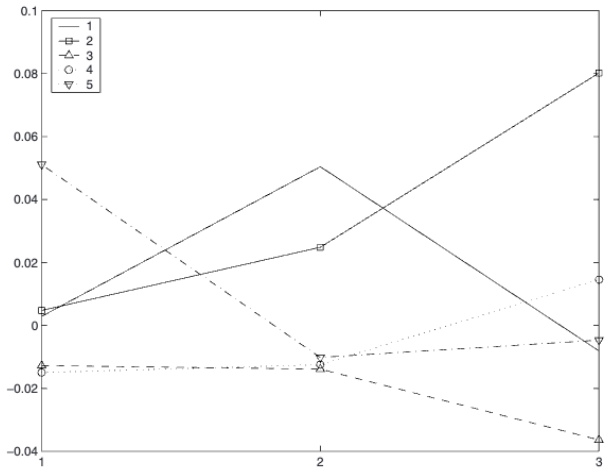


Fig. 2. Centers of five clusters.

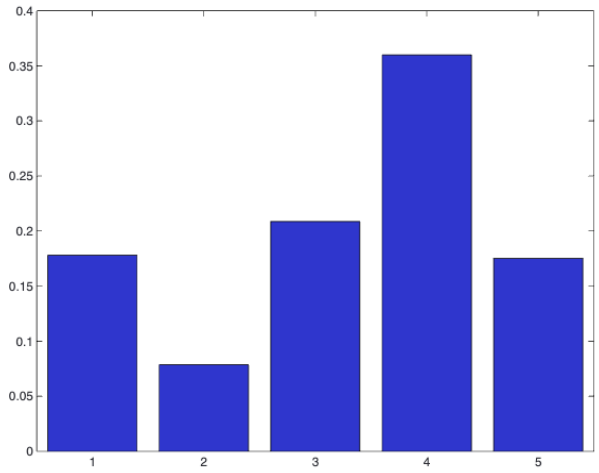


Fig. 3. Priors of five clusters.

parameter estimating

two task

1. task 1 We have only observations and we want to estimate the parameters such as transitional probabilities. we use EM(Expectation maximization) alogrithm to estimate the parameters.
2. task 2 We have observations and parameters, we want to know what's the best possible hidden states. We use a dynamic programming algorithm Viterbi.

To train our HHMM model, we adapt a semi-supervised learning techniques.Rather than learning HHMM from only observed RC sequences, we manually labelled the hidden states in the first level. We consider in this paper the trend as the interval to which the gradient of moving average of future price including current price belongs. Moving average (MA) is defined as:

$$MA(i)^W = \sum_{t=0}^{W-1} \text{close}(i+t)$$

where i is the i-th day and W is the length of window to be considered. The trend of i-day is labelled to one of five hidden states according to the gradient of the moving average line. Using the labelled data, the HHMM is trained by EM algorithm for dynamic Bayesian Networks.

semi-supervised learning

Combination of Labeled and Unlabeled Data: Semi-supervised learning uses both labeled and unlabeled data. The labeled data is used as in supervised learning, while the unlabeled data is handled using techniques from unsupervised learning (like the Baum-Welch algorithm).

Parameter Initialization: The parameters (transition, emission, and initial state probabilities) are first initialized using the labeled data, following the same process as in supervised learning.

Experiment

The trading strategy

Online Learning: In some applications, this parameter updating can be done in real-time or online, allowing the HMM to adapt continuously as new data comes in. This is common in applications like speech recognition, where the model needs to adapt to different speakers or accents.

Updating the transitional parameter when receiving the data stream.

We take a bid signal if the sum of probabilities of su and wu gets greater than 0.5. A ask signal if the sum for sb and wb gets lower than 0.5.

	probability > 0.5	**market probability < 0.5**
p(su)+p(wu)	bid	hold
p(sb)+p(wb)	hold	ask

Table 1. Comparison of the trading performances.

Company	Baseline 1	Baseline 2	HHMM
Samsung Electronics	47.38	14.67	29.78
SK Telecom	-1.39	-4.68	6.33
POSCO	25.87	2.08	18.66
KEPCO	10.96	1.07	4.91
KT	-8.80	-5.91	3.28
Hyundai Motors	52.21	27.61	32.14
LG Electronics	30.00	17.70	19.66
Samsung SDI	75.00	14.95	30.22
Hyundai MOBIS	52.55	49.78	53.83
Shinsegae	44.53	10.02	18.76
KT&G	32.58	16.08	17.12
LG Chemical	19.56	22.80	25.01
DSME	30.85	-2.20	33.12
KOGAS	0.74	-1.24	-0.50
Hankook Tire	74.49	29.35	24.53
NC Soft	13.49	3.76	15.77
Hanlla Climate Control	55.65	16.94	27.01
Cheil Industries	5.71	17.87	9.66
Daum Communication	-32.75	-11.42	-0.71
Handsome	2.91	4.07	21.09

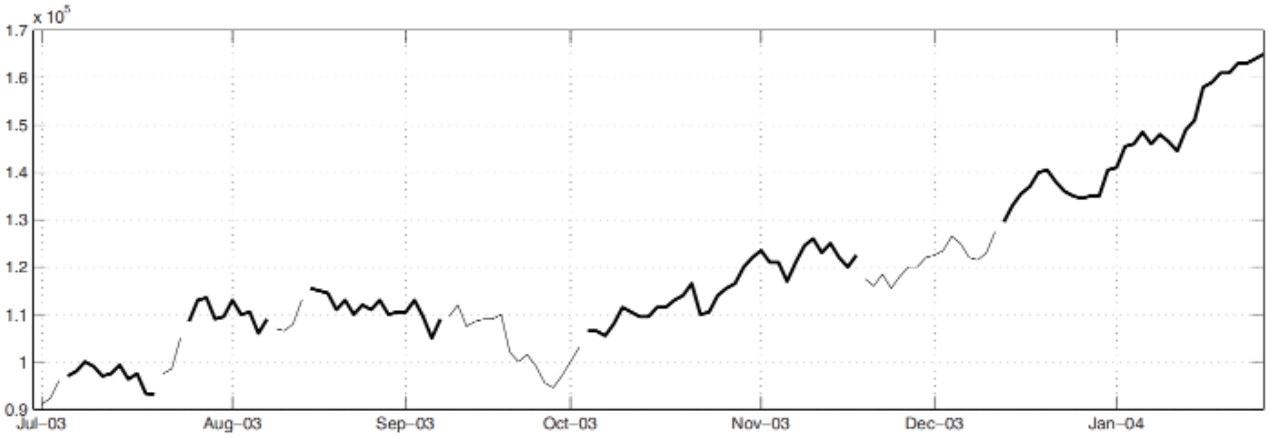


Fig. 4. Simulated trading on Samsung SDI.

Fig. 5. Hinton diagram.

Baseline 1: Imaginary profit as if we would hold a stock during entire test period.

Baseline 2: Cumulative profit when we use TRIX, a technical indicator, of which gold cross is used as bid signal and dead cross is used as ask signal.